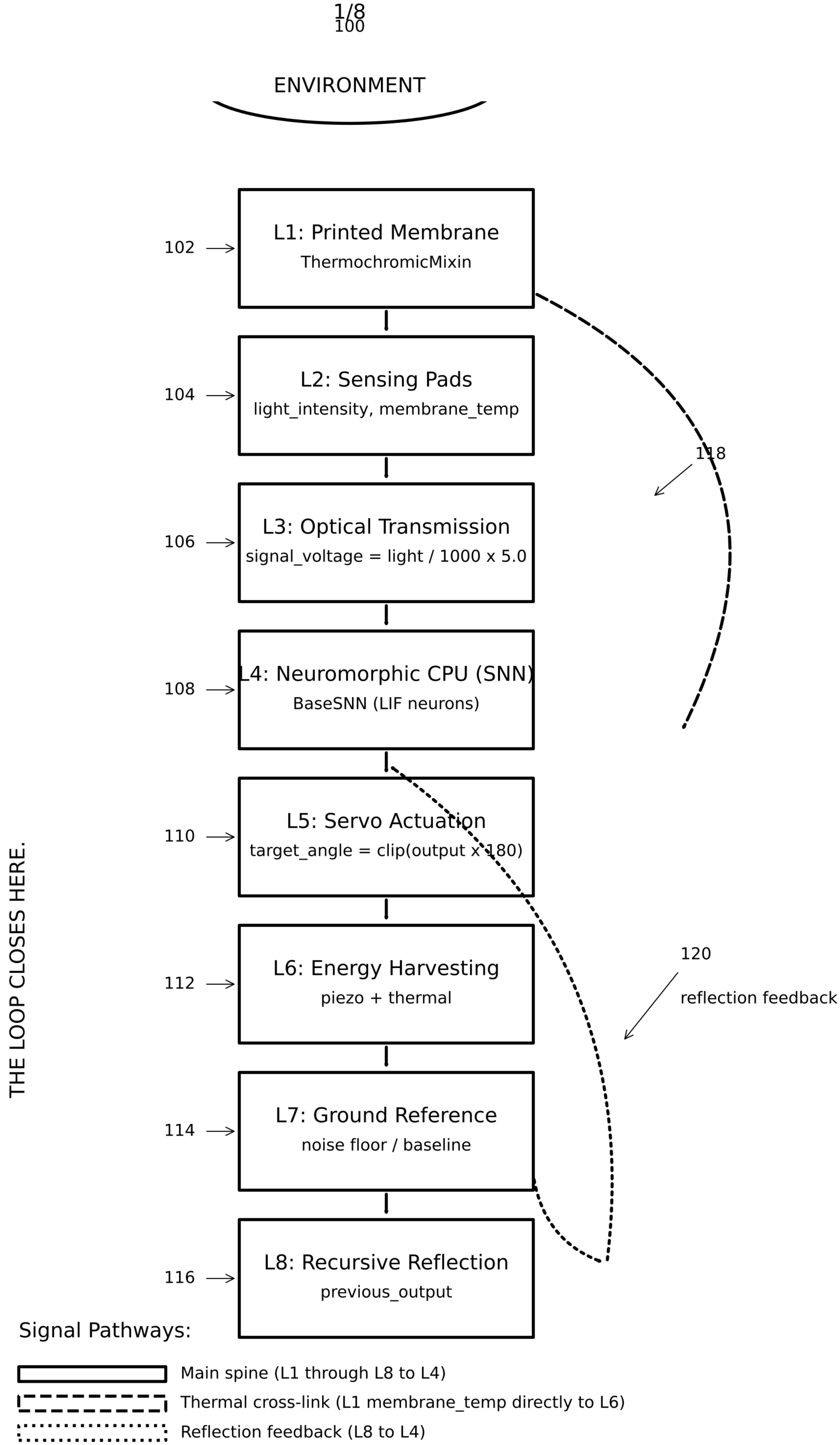

100
ENVIRONMENT
102
L1: Printed Membrane
ThermochromicMixin
104
L2: Sensing Pads
light_intensity, membrane_temp
106
L3: Optical Transmission
signal_voltage = light / 1000 x 5.0
108
L4: Neuromorphic CPU (SNN)
BaseSNN (LIF neurons)
110
L5: Servo Actuation
target_angle = clip(output x 180)
112
L6: Energy Harvesting
piezo + thermal
114
L7: Ground Reference
noise floor / baseline
116
L8: Recursive Reflection
previous_output
118
120
reflection feedback
THE LOOP CLOSES HERE.
Signal Pathways:
Main spine (L1 through L8 to L4)
Thermal cross-link (L1 membrane_temp directly to L6)
Reflection feedback (L8 to L4)

FIG. 1

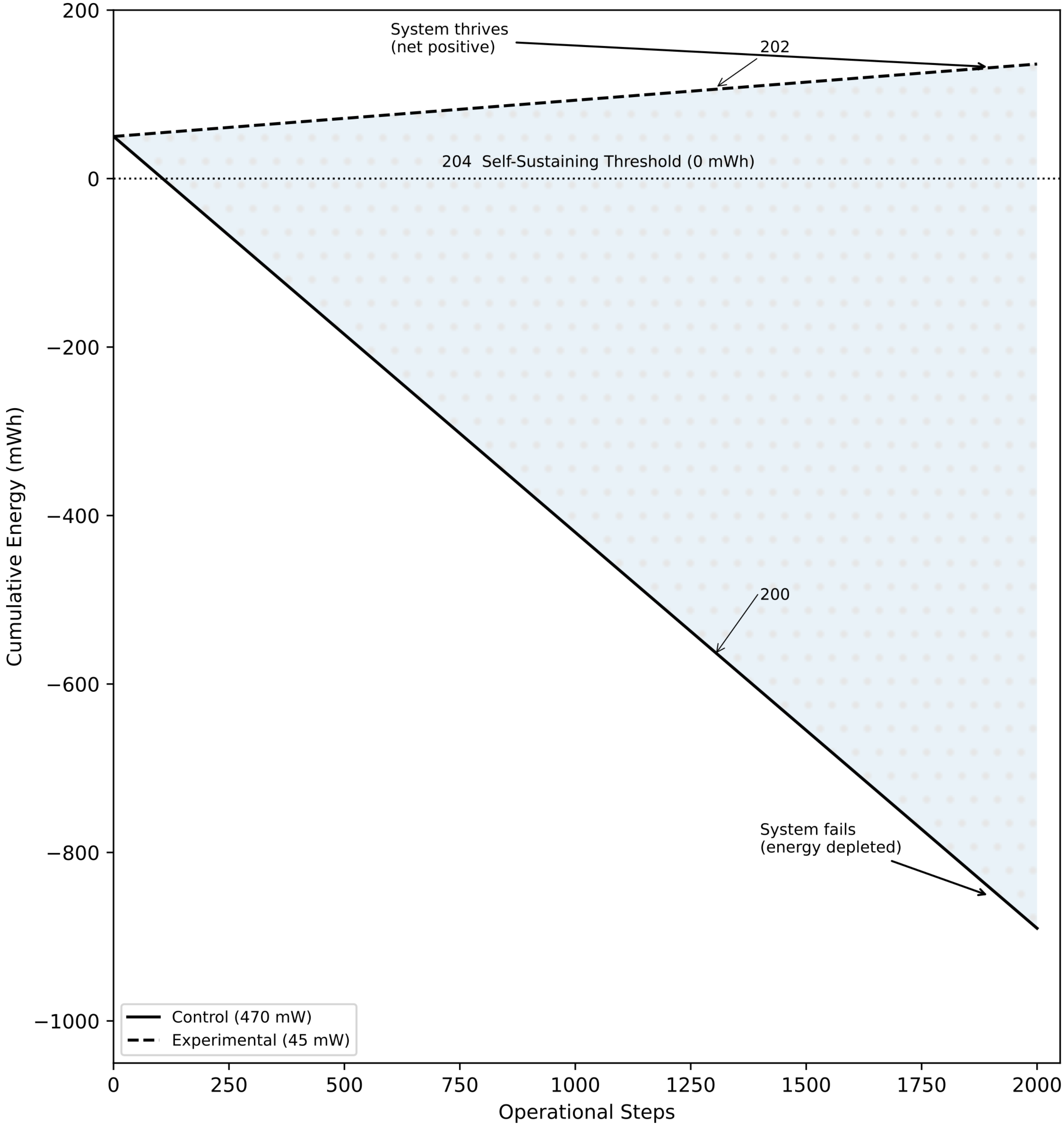

System thrives
(net positive)
202
204 Self-Sustaining Threshold (0 mWh)
200
System fails
(energy depleted)
Control (470 mW)
Experimental (45 mW)
Cumulative Energy (mWh)
Operational Steps
200
0
−200
−400
−600
−800
−1000
0
250
500
750
1000
1250
1500
1750
2000

FIG. 2

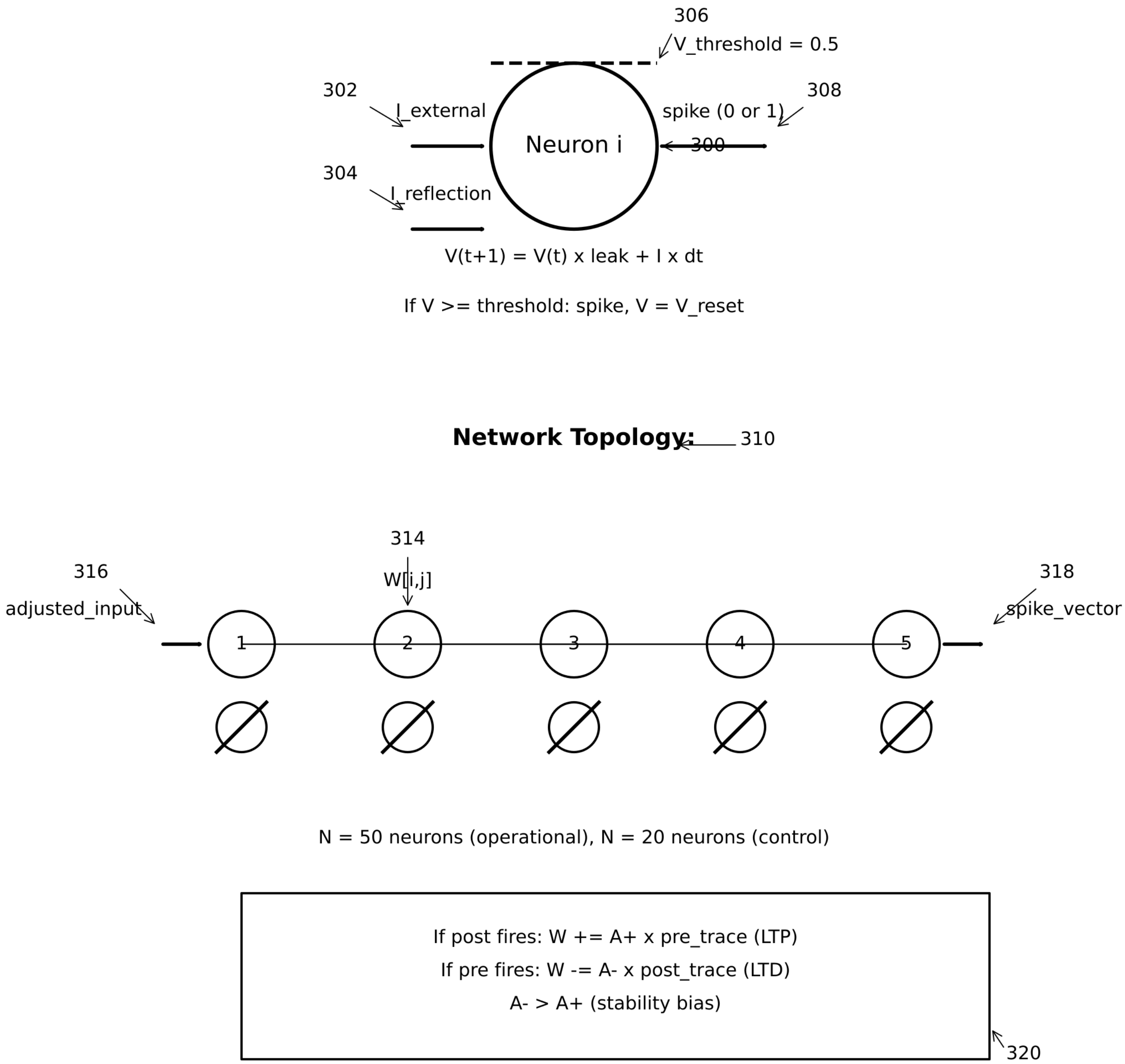

306
V_threshold = 0.5
302
I_external
Neuron i
spike (0 or 1)
308
300
304
I_reflection
V(t+1) = V(t) x leak + I x dt
If V >= threshold: spike, V = V_reset
Network Topology:
310
314
W[i,j]
316
adjusted_input
318
spike_vector
1
2
3
4
5
N = 50 neurons (operational), N = 20 neurons (control)
If post fires: W += A+ x pre_trace (LTP)
If pre fires: W -= A- x post_trace (LTD)
A- > A+ (stability bias)
320

FIG. 3

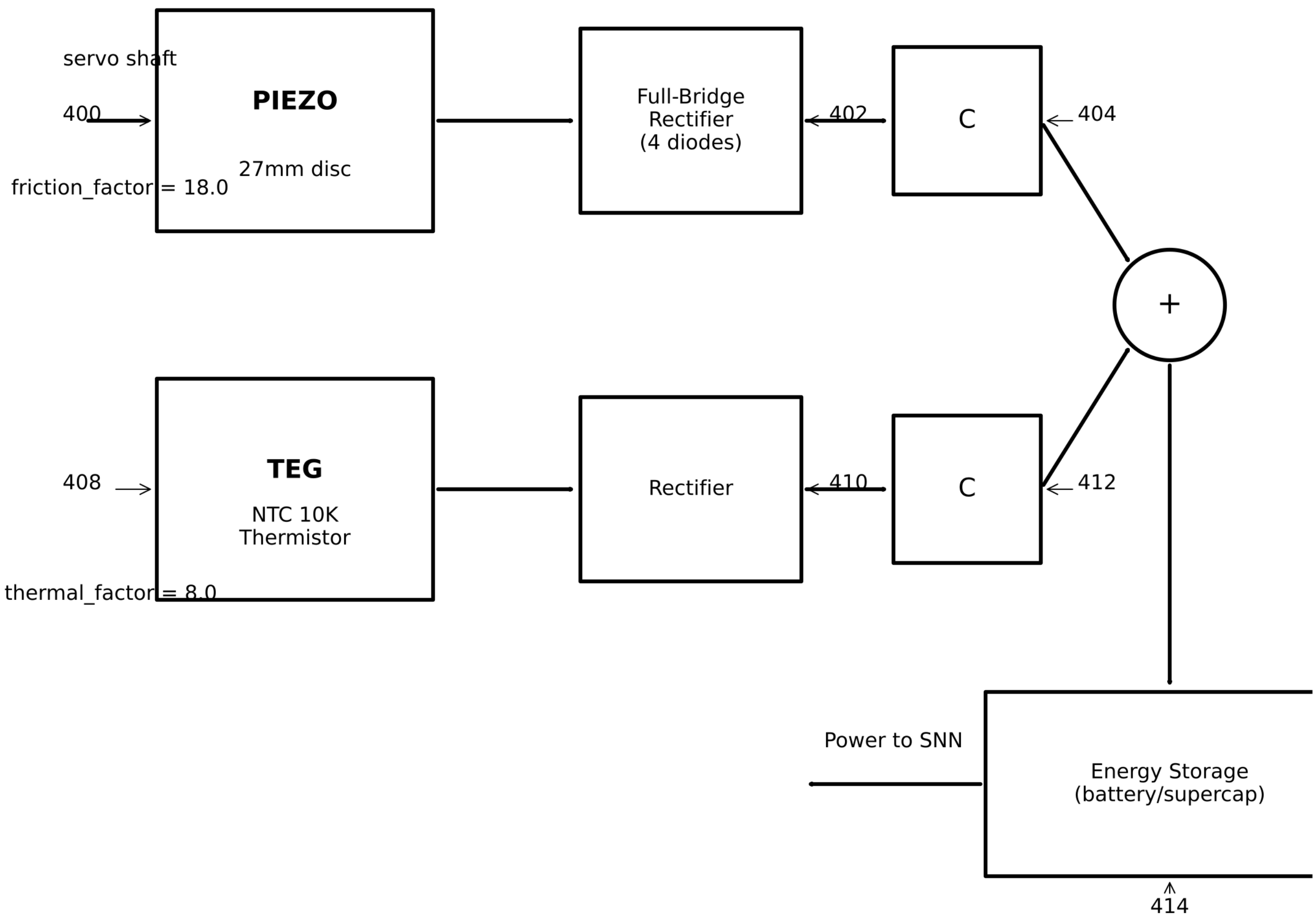

servo shaft
400
friction_factor = 18.0
PIEZO
27mm disc
Full-Bridge
Rectifier
(4 diodes)
402
C
404
+
408
thermal_factor = 8.0
TEG
NTC 10K
Thermistor
Rectifier
410
C
412
Power to SNN
Energy Storage
(battery/supercap)
414

FIG. 4

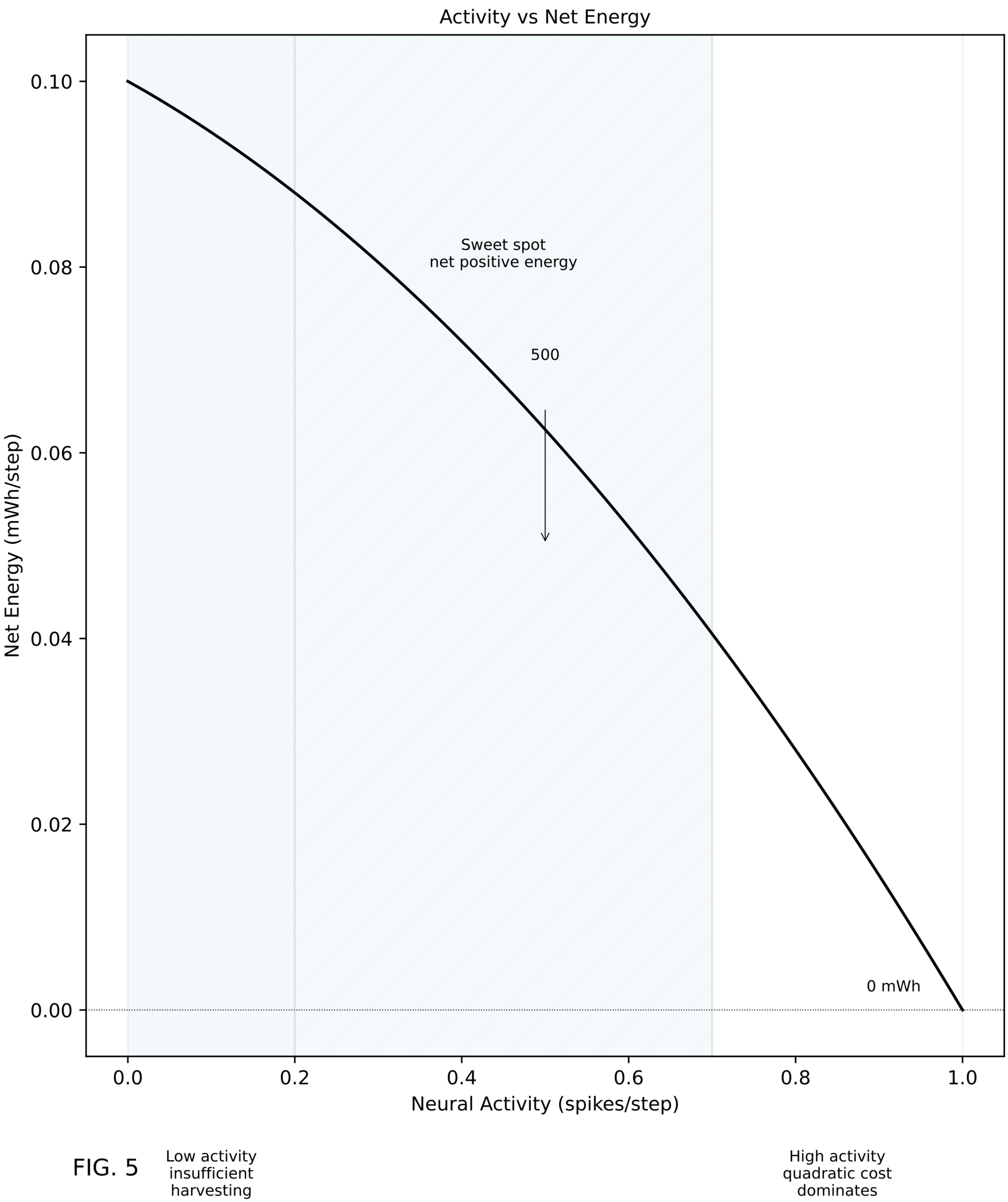
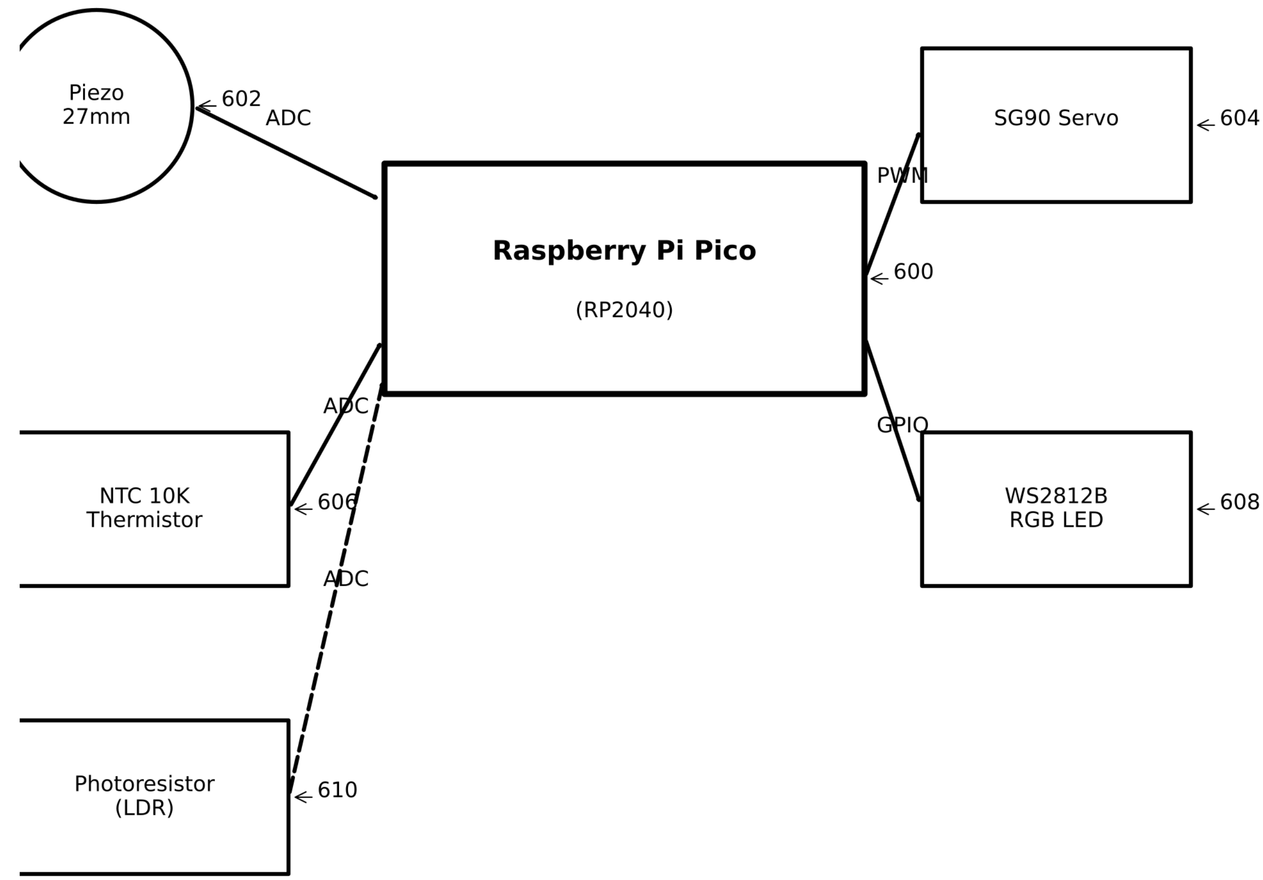

Activity vs Net Energy
Sweet spot
net positive energy
500
0 mWh
0.10
0.08
0.06
0.04
0.02
0.00
Net Energy (mWh/step)
0.0
0.2
0.4
0.6
0.8
1.0
Neural Activity (spikes/step)
Low activity
insufficient
harvesting
High activity
quadratic cost
dominates

FIG. 5



Bill of Materials

Raspberry Pi Pico	$4
SG90 Micro Servo	$3
27mm Piezo Disc	$2
NTC 10K Thermistor	$1
Photoresistor (LDR)	$1
WS2812B RGB LED	$1
Misc (wires, board)	$3

Total BOM Cost: <$25

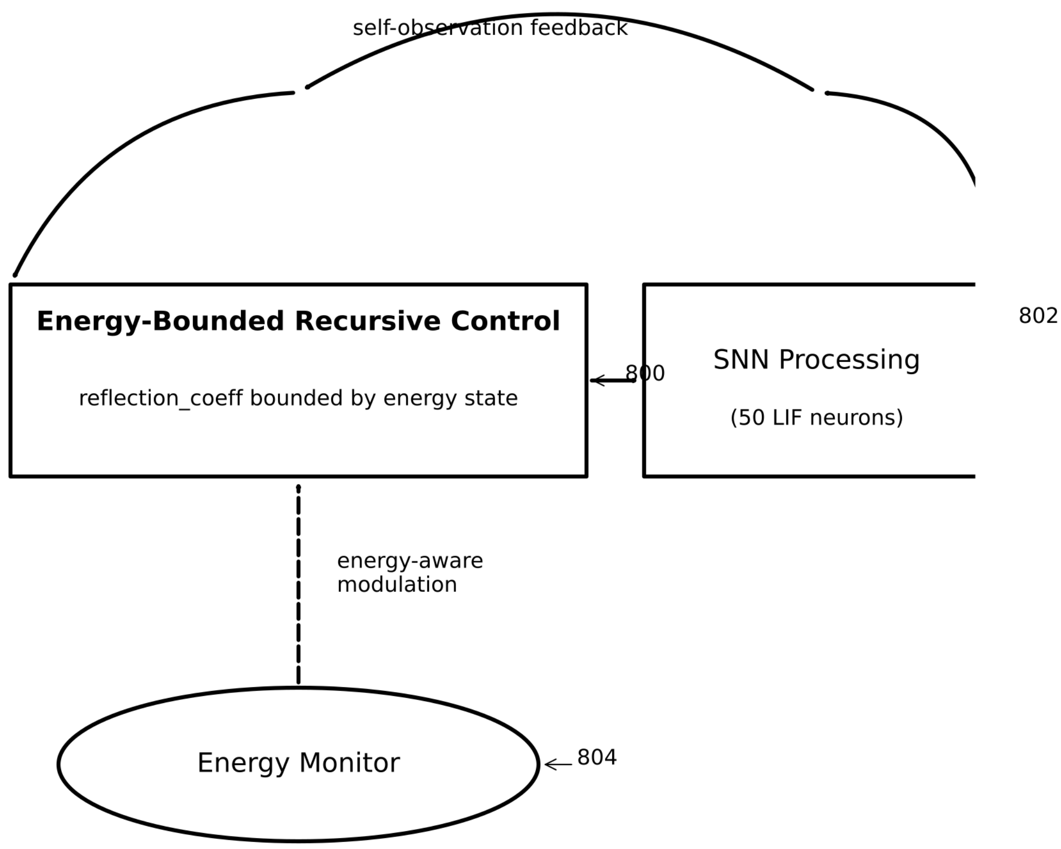
FIG. 6

Claim	Criterion	Measured	Threshold	Result
	Phase 7 Control Baseline (EnergyConfig)			
A	Closed-loop continuity	2000 pts	2000	PASS
B	Boundedness	bounded	bounded	PASS
C	Robustness (noise)	Std(θ)≤45°	≤45°	PASS
D	Input-output gain	corr≥0.2	≥0.2	PASS
E	Saturation ratio	sat≤0.20	≤0.20	PASS
	Phase 7 Full Integration (Control)			
--	Energy at 2000 steps	-4,697 mWh	< 0	CONFIRMED
	MCP Integration (BalancedEnergyConfig)			
--	Energy at 2000 steps	+4,170 mWh	> 0	CONFIRMED
	Phase 8 STDP			
F8.1	Weight entropy decrease	2.95→2.02	decrease	PASS
F8.2	STDP MI > 1.2x frozen	6.52x	> 1.20	PASS
F8.3	Weight convergence	0.00%	< 10%	PASS

700

All claims validated. System reproducible via git tags.

FIG. 7



Energy < 15 mWh: inhibit (mod = -0.4)

Energy < 30 mWh: cautious (mod = -0.1)

Energy > 80 mWh: explore (mod = +0.3)

Recursive control loop bounded by energy levels to ensure self-sustainability.

FIG. 8